

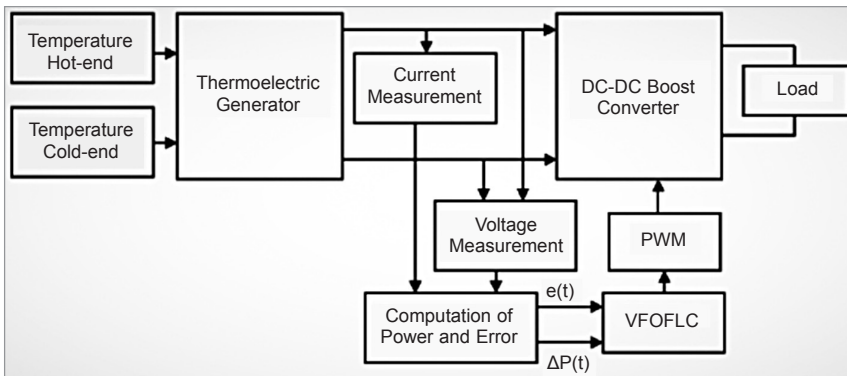


Fig. 10: Block diagram of thermoelectric power harvesting system

TABLE 2
ZT VALUES OF BULK AND NANO-STRUCTURED SI-GE TE MATERIALS

Material	Bulk		Nanostructured	
	ZT max	Temp. (°C)	ZT max	Temp. (°C)
Si ₈₀ Ge ₂₀ (n-type)	1.0	1200	1.3	1173
Si ₈₀ Ge ₂₀ (p-type)	0.7	1200	0.95	1073

300K to 1300K will have an efficiency of about 12.1%.

Block diagram of a typical thermoelectric power harvesting system is shown in Fig. 10.

A thermoelectric power generation system has low energy conversion efficiency. According to the principle of maximum power transfer, the system would transfer the maximum power when the load resistance is equal to the internal resistance of the TEG. Based on this principle, several maximum power point tracking (MPPT) algorithms may be selected with different control logic to harvest the maximum power from the TEG. If a TEG is connected directly to the load, the load impedance will set the operating point that might result in the TEG output being less than the maximum output power.

The MPPT technique is considered an efficient mechanism to improve the performance of TEG by increasing the conversion efficiency. In this system, operating point of the TEG is moved promptly towards an optimal point to increase the energy

harvesting by using a variable fractional order fuzzy logic controller (VFOFLC) based MPPT technique. The variable tracking step size is applied using a dynamic variable fractional factor whose value is calculated based on the voltage output

of the TEG. The fraction order term introduced in the MPPT algorithm would contract or expand the input domain of the fuzzy logic controller to shorten the tracking time and maintain a steady-state output around the maximum power point.

Optimisation of the TEG

It has been found that the output power is correlated with the geometry of the device. By changing the leg height and the number of thermoelectric pellets to an optimum value it is possible to maximise electric power or efficiency at given operating conditions. There is interdependence between optimal leg geometry and the electrical load resistance.

If number of legs is low, the energy conversion is low, because the load resistance (R_L) is not sufficient to obtain an adequate high voltage and vice versa. A reduction of the leg length leads to a reduction of the electrical resistance, and an increase of the leg length leads to the higher temperature difference across the TEG. If the geometric parameters like leg length, number of semicon-

ductor pellets, the base area ratio of the semiconductor columns are optimised, the output power and thermal efficiency are considerably improved.

The shape of the legs of TEG devices has considerable effect on the device performance. The conventional rectangular leg shape found in commercial TEGs is not the optimal shape for heat-to-power energy conversion. The hourglass shaped TE legs result in more than double the electrical potential and maximum power compared to conventional rectangular shape. The trapezoid leg with the largest cross-sectional area at the hot side results in about double the electrical potential and a 50% increase in the power output compared to the conventional rectangular shape. The electrical output power values, if optimised, can be 890% higher than a random value without optimisation.

Heat sink is required at the TEG when a high heat flow rate is applied on the hot side of the TEG. In order to have quick heat dissipation at the cold side cooling radiators are provided there, so that bigger temperature difference across the TEG can be obtained. The fins attached to the heat sinks are very important for enhancing the heat transfer at the hot and cold sides. The heat transfer increases when the number of fins are increased and the fin height is more, due to more heat transfer area. However, increase of heat transfer area is limited to an optimum value beyond which the change in the output electrical power becomes less significant.

The thermal resistance of the heat sink (R_{hs}) is:

$$R_{hs} = (T_{\text{heat sink}} - T_{\text{amb}}) / Q_h$$

where, $T_{\text{heat sink}}$ is the sink temperature, T_{amb} is the environmental temperature, and Q_h is the heat flow. Experimental results show that increasing the thermal resistance of both cold and hot side heat sinks by 10% improves the electrical output power by 8%.